INFT6201 — Data Science Project

Who Gets Hurt, Where and Why?

Severity analytics on New South Wales road crash data, 2020–2024

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Code and reproducible outputs: https://github.com/QuangThanhDS/nsw-road-crash-analytics

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Executive summary

New South Wales (NSW) recorded 340 road deaths in 2024, and the State's 2026 Road Safety Action Plan commits to halving deaths and cutting serious injuries by 30% by 2030 (Transport for NSW, 2022, 2025a). This report uses the Transport for NSW open crash data for 2020–2024 (Transport for NSW, 2025b), which covers 92,082 police-reported crashes and 1,507 deaths, to ask which crash circumstances turn a crash into a fatal or serious one, and where those crashes happen.

Three findings stand out from the preliminary analysis:

- **Regional rural roads carry a disproportionate share of trauma.** They account for 12.7% of crashes but 39.2% of deaths. At the same posted speed limit, a crash outside Greater Sydney–Newcastle–Wollongong is roughly two to four times more likely to be fatal (5.0% vs 1.4% at 100 km/h).
- **Who is involved matters more than when or in what weather.** Crashes involving a motorcycle, pedestrian or cyclist, or a collision with a tree or pole, carry the highest odds of a killed or seriously injured (KSI) outcome. Weather and day of week add almost nothing.
- **Data quality can mislead a model.** The *Street lighting* field is left "unknown" far more often for minor crashes than for fatal ones, which leaks the outcome into the model. Removing it lowers the apparent ROC-AUC from 0.81 to a more honest 0.75.

A temporally validated gradient-boosting model ranks 2024 crashes well (the top risk decile has a 61% KSI rate against 5.5% in the bottom decile), and an interpretable logistic regression performs almost as well. That makes logistic regression the better fit for public-sector stakeholders who need to explain their decisions.

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1 Case study

1.1 Context

Road trauma is a leading cause of preventable death worldwide (World Health Organization, 2023). In NSW, more than two-thirds of road deaths occur on country roads, even though country residents make up about one-third of the population (Transport for NSW, 2022). The 2026 Road Safety Action Plan sets targets of a 50% reduction in deaths and a 30% reduction in serious injuries from the 2018–2020 baseline by 2030 (Transport for NSW, 2022). Meeting these targets needs evidence on *which* crashes become severe, so that limited infrastructure, enforcement and education budgets can go where they prevent the most trauma.

1.2 Application examples

The analysis is designed to support decisions such as:

- **Safer-roads investment:** prioritising route segments with clusters of severe run-off-road and head-on crashes for barriers, rumble strips or tree and pole treatments.
- **Vulnerable road user programs:** targeting motorcycle, pedestrian and cyclist safety where these users make up a large share of KSI crashes.
- **Enforcement and patrol scheduling:** aligning police resources with the times and places where crashes are most likely to be severe, not just most frequent.
- **Speed-zone review:** using severity differences across speed limits and regions to inform reviews of 80–110 km/h rural zones.

1.3 Data science questions

Q1. Severity factors: Which road, environment and road-user characteristics are associated with a reported crash being fatal or serious (KSI), and how well can they predict it?

Q2. Regional gap: Why are regional crashes more severe than metropolitan ones, and does the gap persist after accounting for speed limit and crash type?

Q3. Hotspots: Where are crashes and severe crashes concentrated, and which local government areas (LGAs) carry the largest burden?

Q4. Temporal patterns: How do volume and severity vary by year (including the COVID-19 period), day of week and time of day?

This preliminary report answers Q1 with initial models and gives descriptive evidence for Q2–Q4.

1.4 Stakeholders

Table 1 lists the main stakeholders and how each would use the results.

Table 1: Key stakeholders and their interest in the analysis.

Stakeholder	Interest and use of results
Transport for NSW, Centre for Road Safety	Data custodian and policy owner; program targeting, progress against 2030 targets
Local councils	Local road treatments, 40 km/h zones, LGA-level benchmarking
NSW Police Force	Timing and location of enforcement (e.g. weekend mornings, regional highways)
NSW Health and emergency services	Regional trauma response times and hospital demand
Insurers (icare, CTP insurers)	Injury burden and claims risk
Road user advocacy groups	Evidence for motorcycle, cycling and pedestrian safety
General public	Transparent communication of risk

2 Datasets

2.1 Background

The primary source is the *NSW Road Crash Data 2020–2024* release published by Transport for NSW on its Open Data Hub under a Creative Commons Attribution licence (Transport for NSW, 2025b). The release is updated annually as a rolling five-year window; the 2020–2024 release (updated 9 October 2025) is the most recent available. It contains crashes that meet national reporting criteria: the crash was reported to police, occurred on a public road, involved at least one moving vehicle, and resulted in a death, an injury or a vehicle being towed away (Transport for NSW, 2021).

The data comes as two linked tables (Table 2). The *Data Manual* defines every field (Transport for NSW, 2021). Severity levels below "fatal" come from linking police reports with health records: a *serious injury* crash is one where at least one person was matched to a hospital admission on the day of the crash or the day after, while a *moderate injury* is matched only to an emergency department attendance. Fatalities are deaths within 30 days of the crash.

To map the data, simplified Australian Statistical Geography Standard (ASGS) Edition 3 boundaries for the state, Greater Sydney and 2024 LGAs were used (Australian Bureau of Statistics, 2021). They were obtained in converted form from the `nswgeo` package (Suster, 2025).

Table 2: Datasets used in the analysis.

Table	Size	Content
CRASH	92,189 rows × 50 columns	One row per crash: year, month, weekday, two-hour time band, location (coordinates, LGA, suburb, street), road and environment (speed limit, alignment, surface, weather, lighting), crash type (RUM/DCA codes, first impact type), key vehicle type, and counts of people killed and injured by severity.
TRAFFIC UNIT	170,962 rows × 10 columns	One row per traffic unit (vehicle, pedestrian, cyclist): unit type, role in first impact, direction, manoeuvre and objects hit. Linked to CRASH by `Crash ID` (one-to-many).
ASGS boundaries	1 state, 1 GCCSA, 129 LGAs	Polygon boundaries (GDA2020) used for mapping.

2.2 Data quality issues

A systematic profile of all 50 CRASH fields (missing values, "unknown"-coded values, cardinality and range checks) and a referential-integrity check between tables found the issues summarised in Table 3. The structure of the data is clean: there are no duplicate crash IDs and no orphan traffic units, and every crash has at least one unit. The problems lie in *what was recorded and how*.

Table 3: Data quality issues identified and how they were handled.

Issue	Evidence	Handling
Outcome-dependent recording (target leakage)	*Street lighting* is "Unknown / not stated" for 36.5% of crashes overall, but for 62.8% of tow-away and 52.9% of minor-injury crashes compared with 0.6% of fatal ones. The KSI rate is 3.6% when the field is unknown and 26–38% otherwise.	Excluded from the main model; a comparison run quantifies the leakage (Section 5).
Structural missingness	*Primary temporary / hazardous / permanent feature* are 98.5%, 98.0% and 73.0% blank; *Other TU type* is 30.4% blank (single-vehicle crashes). A blank means "none", not "unknown".	Not used yet; to be recoded as "None".
Coverage and under-reporting	Only police-reported injury or tow-away crashes are included; minor crashes without a tow-away are absent.	Results describe severity *given* a reported crash, not crash risk per trip.
Late reporting	107 crashes from 2019 appear in reporting year 2020, and 63–167 crashes each year are reported in the following year.	Analysis uses *Year of crash* and keeps 2020–2024 only (92,082 crashes).
Consistency with official totals	Deaths in the data match the official 2023 toll (340) exactly, but 2024 shows 327 against an official 340 (Transport for NSW, 2025a).	Noted; 2024 counts may be revised in later releases.
Definitional overlap	*Urbanisation* is itself derived from LGA and speed limit (country urban $\leq$80 km/h, non-urban >80 km/h; Transport for NSW, 2021).	Regions are compared as metro vs rest of NSW *within* speed limits.
Coarse time and privacy limits	Time is given only as two-hour bands, month and weekday. No driver age, licence, speeding, alcohol or restraint fields are released.	Time treated as categorical; human factors listed as a limitation.
Spatial outliers	Five crashes at 159°E look anomalous but are on Lord Howe Island (part of NSW).	Kept; excluded only from the mainland map.

The first issue is the most consequential. Police record crash circumstances in more detail when someone is badly hurt. As a result, whether a field was *filled in* carries information about the outcome, which is a classic form of leakage (Kaufman et al., 2012). Smaller "unknown" categories in *Natural lighting* and *Key TU type* (each under 2.5% of crashes) show the same pattern and are flagged for treatment in the next stage.

3 Data modelling

3.1 Conceptual data model

The two source tables form a one-to-many relational model (Figure 1). For analysis, the TRAFFIC UNIT table was aggregated to the crash level as binary flags (for example *any pedestrian involved*,

any motorcycle involved, *tree or pole hit*) and joined to CRASH. This gives one row per crash, which matches the unit of the severity outcome.

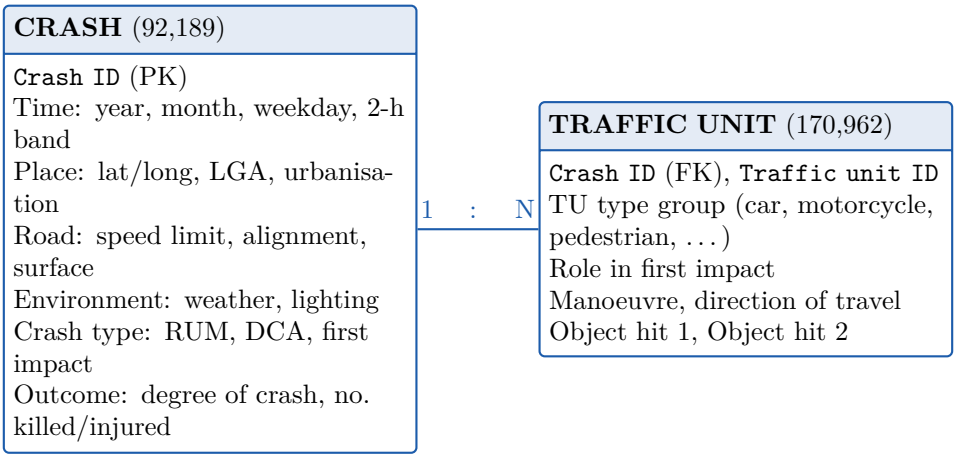


Figure 1: Relational model of the TfNSW crash data. Traffic units are aggregated to crash-level flags before modelling.

3.2 Analytical problem formulation

Crash severity is an ordered, five-level outcome, but its two extremes are very unbalanced (1.5% fatal against 32.0% non-casualty). The preliminary model therefore uses a binary target that matches the policy target: **KSI**, which equals 1 if the crash was fatal or serious-injury (22.8% of crashes). Predictors were limited to circumstances recorded at the scene, giving 16 categorical crash attributes and 7 crash-level flags from the TRAFFIC UNIT table. Injury counts were excluded because they define the target, and *Street lighting* was excluded because of the leakage described above.

3.3 Models

- **Majority-class baseline**: always predicts "not KSI", to anchor the metrics.
- **Logistic regression** (L2-regularised, one-hot encoded, balanced class weights). This is the standard interpretable model in crash-severity research (Savolainen et al., 2011); its coefficients convert to odds ratios that stakeholders can read directly.
- **Histogram gradient boosting** (native categorical handling, balanced class weights), a tree ensemble in the gradient-boosting family (Friedman, 2001) that uses LightGBM-style histogram binning (Ke et al., 2017). It captures non-linear effects and interactions such as speed × region without having to specify them.

Models were implemented with scikit-learn (Pedregosa et al., 2011), and data handling used pandas (McKinney, 2010).

3.4 Validation design

A **temporal split** was used: models were trained on 2020–2023 (73,143 crashes) and tested on 2024 (18,939 crashes). This mirrors the intended use, learning from past years to flag high-risk crash patterns in the next year, and it is stricter than a random split because it exposes the models to year-to-year drift. Because the classes are unbalanced, accuracy is not reported. Evaluation

uses ROC-AUC, precision–recall AUC (which is more informative under class imbalance; Saito and Rehmsmeier, 2015), KSI recall and precision, and the share of fatal crashes flagged as high risk (He & Garcia, 2009). Feature influence is assessed through odds ratios (logistic regression) and permutation importance (gradient boosting), measured as the drop in test ROC-AUC when a feature is shuffled (Breiman, 2001).

3.5 Suitability for the case study

- **Logistic regression** is the most suitable *primary* model for this case study. The stakeholders are public agencies that must justify their investment decisions, and odds ratios translate directly into program priorities. It performs only 0.007 AUC below gradient boosting (Section 5).
- **Gradient boosting** works best as a *check* on the logistic model. Its small gain suggests that severity is driven mostly by additive main effects in this feature set. Its importance ranking independently confirms the logistic model's drivers.
- **Limitations of both:** predicted probabilities are inflated by class weighting and need recalibration before they are reported as risks (Niculescu-Mizil & Caruana, 2005). Neither model accounts for the unobserved heterogeneity that the accident-research literature stresses. Ordered and random-parameters models are the natural extension for the full five-level severity scale (Mannering & Bhat, 2014; Savolainen et al., 2011).
- **Spatial models** (density-based clustering of KSI crashes) are better suited than classifiers to Q3 and are planned for the next stage.

4 Data visualisation and communication of results

Each visual was chosen to answer one question for one audience, and all were produced with Matplotlib (Hunter, 2007) using a single consistent design system. Table 4 maps each visual to its question and audience.

Table 4: Visualisations, the questions they answer and their intended audience.

Visual (technique)	Q	Purpose	Audience
Severity bar chart (Fig. 2)	–	Class balance; justifies the KSI target	Analysts
Yearly trend lines (Fig. 3)	Q4	COVID dip and post-2022 rise	TfNSW, public
Day × time heatmaps (Fig. 4)	Q4	Volume against severity by time	Police
Speed limit × region lines (Fig. 5)	Q2	Regional gap at equal speed	TfNSW, councils
Impact-type bar chart (Fig. 6)	Q1	Severity by crash mechanism	All
Hexbin maps (Fig. 7)	Q3	Where crashes and severe crashes are	Councils, TfNSW
ROC/PR curves (Fig. 8)	Q1	Model performance	Analysts
Odds ratios and importance (Figs. 9, 10)	Q1	Drivers of severity	Policy makers

Key design choices:

- **Rates alongside counts.** Volume and severity often point in different directions. The heatmaps (Fig. 4) and maps (Fig. 7) pair a count view with a KSI-share view, so that "where crashes are common" is not mistaken for "where crashes are dangerous".

- **Hexagonal binning for 92,000 points.** Plotting each crash as a point would create heavy overplotting along the coast. Hexbins aggregate points into equal-area cells (Carr et al., 1987). A logarithmic colour scale keeps sparse regional cells visible next to Sydney cells with thousands of crashes. Cells with fewer than 15 crashes are hidden on the KSI-share map to avoid unstable rates.
- **Geographic context.** ABS state, Greater Sydney and LGA boundaries (Australian Bureau of Statistics, 2021), plus labelled markers for Sydney, Newcastle and Wollongong, let non-technical readers orient themselves.
- **Accessible colour.** Single-hue sequential scales encode magnitude. Categorical series use a palette checked for colour-vision-deficiency separation, and series are also labelled directly or with a legend, so colour is never the only cue.
- **Small-sample suppression and annotation.** Groups with fewer than 100 crashes are hidden in the speed chart, and bar charts are annotated with sample sizes.
- **Models explained in policy terms.** Odds ratios are shown on a log scale centred at 1, so effects that raise and lower risk appear symmetrically.

5 Preliminary results

5.1 Data characterisation and distributions

Severity. Of the 92,082 crashes from 2020 to 2024, 32.0% were non-casualty tow-aways, 28.4% moderate injury, 16.8% minor injury, 21.3% serious injury and 1.5% fatal (1,388 crashes, 1,507 deaths; Figure 2). The KSI target therefore has a prevalence of 22.8%. Most crashes involve two traffic units (59%), and 29% are single-vehicle crashes.

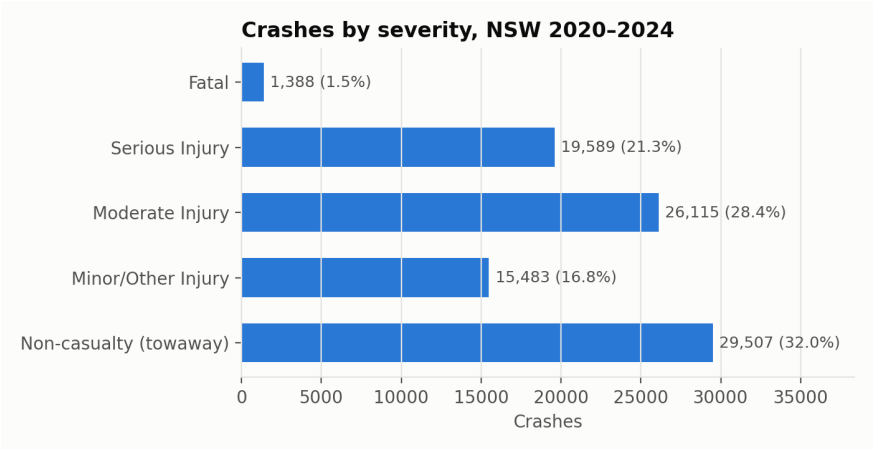


Figure 2: Distribution of crashes by severity, NSW 2020–2024.

Trend. Fatal crashes held at around 260 a year in 2020–2022 and rose to around 300 in 2023–2024. Serious-injury crashes fell in 2021, during COVID-19 lockdowns, and then rose 37% to 4,459 in 2024, while moderate, minor and tow-away crashes stayed flat (Figure 3). Severe crashes are therefore increasing, not crashes overall, which moves NSW away from its 2030 targets.

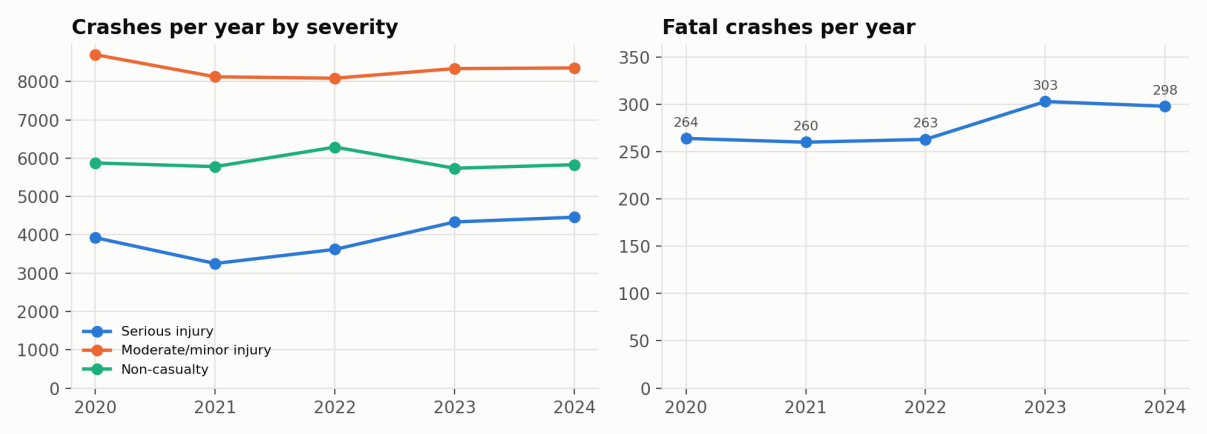


Figure 3: Crashes per year by severity (left) and fatal crashes per year (right).

Time of day and week. Crash volume peaks on weekday afternoons; Friday 16:00–17:59 is the busiest slot, with 2,304 crashes. Severity follows a different pattern: the highest KSI shares occur on *weekend mornings*, reaching 31% on Sunday 10:00–11:59 and 29% on Sunday 08:00–09:59, against about 20% in the weekday peak (Figure 4). Congested peak traffic produces many low-speed crashes, while lighter weekend traffic, often on recreational and regional routes, produces fewer but more severe ones.

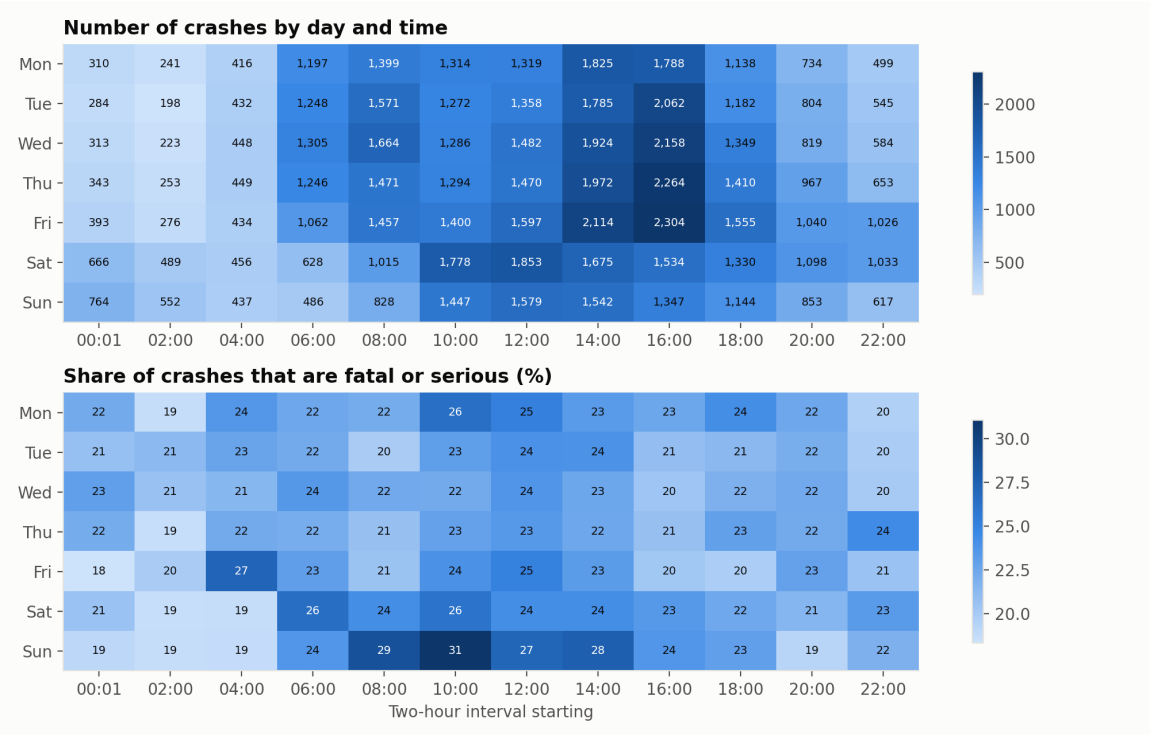


Figure 4: Crash counts (top) and share of crashes that are fatal or serious (bottom) by day and two-hour interval.

Regional gap (Q2). Table 5 shows that regional rural NSW has 12.7% of crashes but 39.2% of deaths, with a fatal-crash rate five times that of the metropolitan conurbation. This matches TfNSW's statement that more than two-thirds of deaths occur on country roads (Transport for NSW, 2022). Even at the same speed limit, crashes outside the conurbation are more severe (Figure 5): at 100 km/h, 5.0% are fatal against 1.4% in the metro areas, and at 80 km/h, 3.4%

against 1.6%. Crash mechanism contributes to the gap: 74.8% of rural crashes are single-vehicle crashes (metro: 18.2%), and 45.1% are vehicle–object impacts (metro: 14.1%).

Table 5: Crashes, deaths and severity by region, 2020–2024.

Region	Crashes	Share of crashes	Deaths	Share of deaths	Fatal rate
Greater Sydney / Newcastle / Wollongong	62,551	67.9%	580	38.5%	0.9%
Regional urban	17,821	19.4%	337	22.4%	1.7%
Regional rural	11,690	12.7%	590	39.2%	4.5%

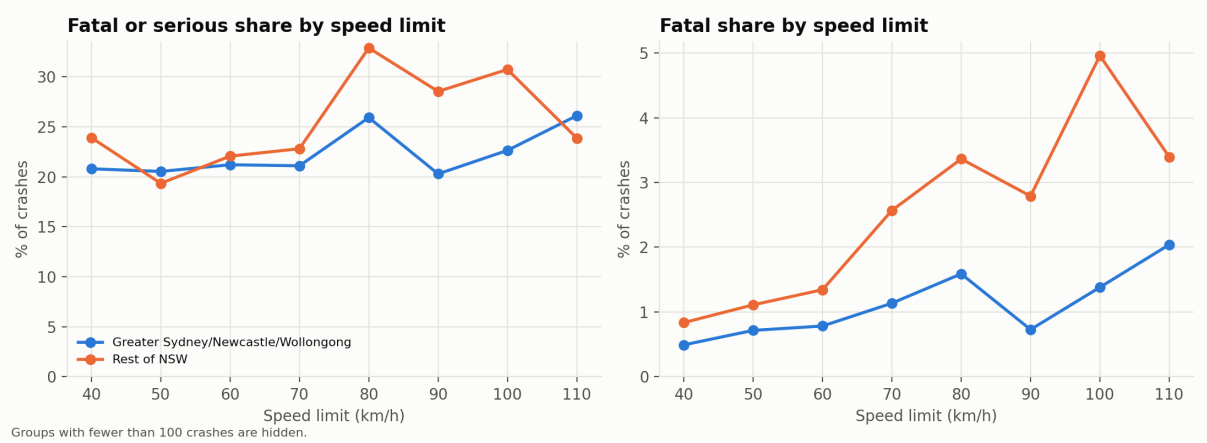


Figure 5: KSI share (left) and fatal share (right) by speed limit, metropolitan conurbation against the rest of NSW.

Crash mechanism. Pedestrian (47% KSI), person–object (46%), head-on (39%) and rollover (32%) crashes are the most severe, and rear-end crashes the least (13%; Figure 6). Head-on crashes have the highest fatal share, at 6.4%.

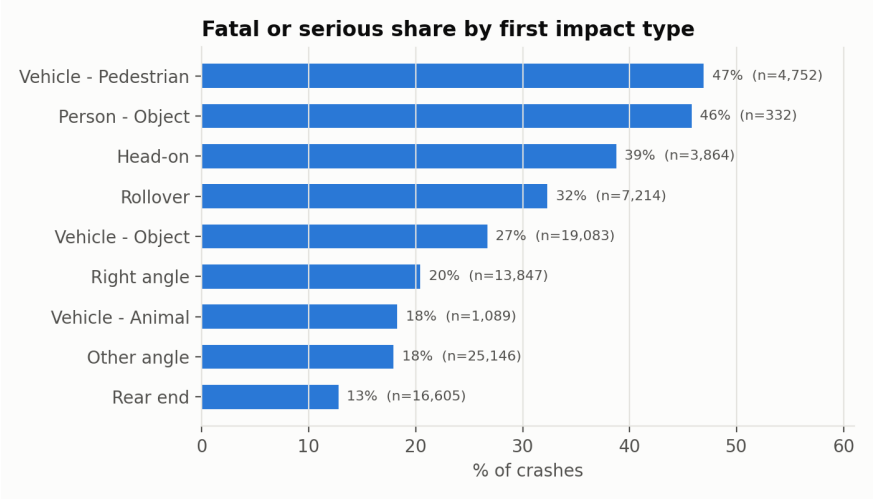


Figure 6: Share of crashes that are fatal or serious, by first impact type (types with at least 100 crashes).

Spatial distribution (Q3). Crashes follow the population and the highway network, with the densest cells in Sydney and along the Pacific, Hume and New England corridors (Figure 7,

left). Within Greater Sydney, KSI shares appear higher on the urban fringe and along major arterials than in the dense inner suburbs (Figure 7, right). By count, the LGAs with the most KSI crashes are Central Coast (1,068), Canterbury-Bankstown (990) and Blacktown (823).

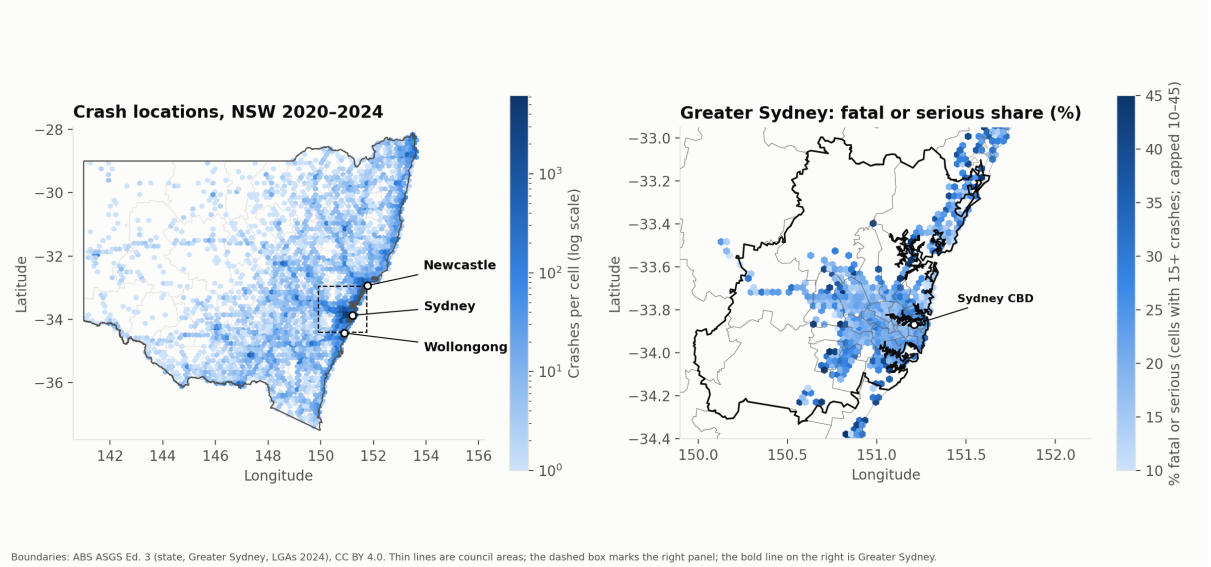


Figure 7: Crash density across NSW, log scale (left), and KSI share within Greater Sydney (right). Boundaries: ABS ASGS Edition 3.

5.2 Initial modelling results

Table 6 reports test-set performance on 2024 crashes. Both models clearly outperform the baseline. Gradient boosting achieves a ROC-AUC of 0.749 and a PR-AUC of 0.498, double the 0.251 that random ranking would give. Logistic regression is within 0.007 AUC. At the default threshold, the models identify about two-thirds of KSI crashes and 85% of fatal crashes. The ROC and PR curves almost coincide (Figure 8).

Table 6: Model performance on the 2024 test set (18,939 crashes, 25.1% KSI).

Model	ROC-AUC	PR-AUC	KSI recall	KSI precision	Fatal crashes flagged
Majority-class baseline	0.500	0.251	0.00	–	0%
Logistic regression	0.742	0.483	0.65	0.44	85%
Gradient boosting	**0.749**	**0.498**	0.67	0.43	85%
Gradient boosting incl. Street lighting (leaky)	*0.809*	*0.553*	*0.81*	*0.45*	*95%*

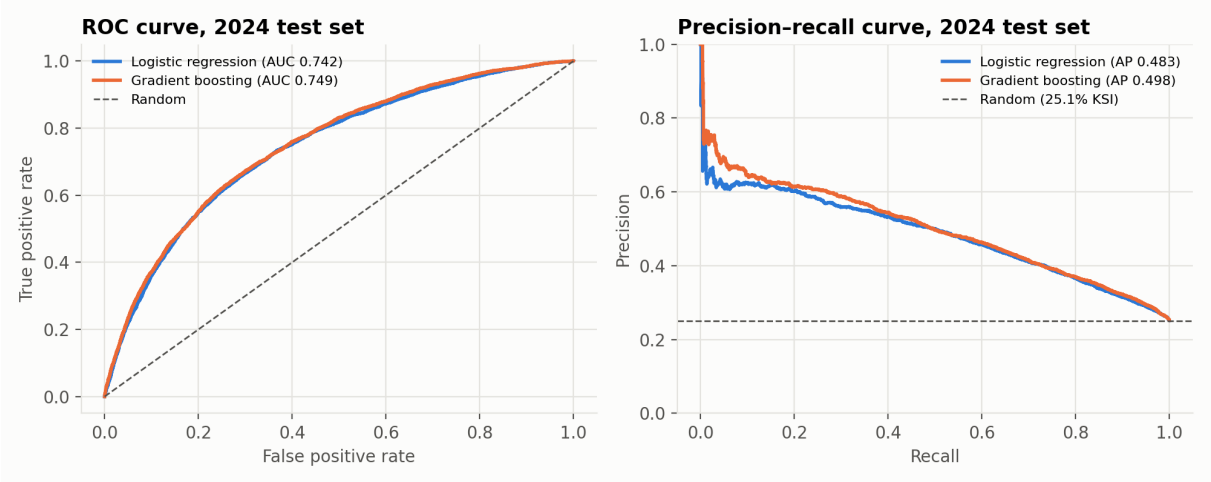


Figure 8: ROC (left) and precision–recall (right) curves on the 2024 test set.

Risk ranking. When 2024 crashes are sorted into deciles by predicted risk, the actual KSI rate rises steadily from 5.5% in the lowest decile to 60.7% in the highest, an eleven-fold spread. However, the top decile's mean predicted probability is 0.79, which confirms the over-estimation expected from class weighting and the need for recalibration.

Leakage check. Adding *Street lighting* raises ROC-AUC to 0.809 and fatal-crash recall to 95% (Table 6, last row). In that run, *Street lighting* becomes by far the most important feature, which only makes sense as a recording artefact. This demonstrates how the data-quality issue in Section 2 would inflate apparent performance, and why the leaky feature was removed.

Drivers of severity (Q1). The two models agree on the main drivers (Figures 9 and 10):

- **Motorcycle involvement** is the strongest factor (odds ratio ≈5.7; largest permutation importance). 49% of motorcycle crashes are KSI, compared with 19% of other crashes.
- **Vulnerable road users**: pedestrian (≈4.4), cyclist (≈4.3) and person–object (≈4.4) involvement, and motorised mobility scooters (≈7.2, though based on few cases).
- **Roadside objects and crash mechanism**: hitting a tree or pole (≈2.5), head-on impact (≈2.0) and heavy-vehicle involvement (≈1.8).
- **Speed environment**: 100–110 km/h zones (≈1.5–1.7). Rear-end crashes (≈0.46), animal strikes (≈0.42) and 40 km/h zones (≈0.53) are associated with lower severity.
- Weather, day of week and school-zone status contribute almost nothing once these factors are known.

Remaining "unknown" categories (for example *Natural lighting: Unknown*, odds ratio ≈0.17) show the same recording effect and will be handled in the next stage. These are associations and should not be read as causal effects.

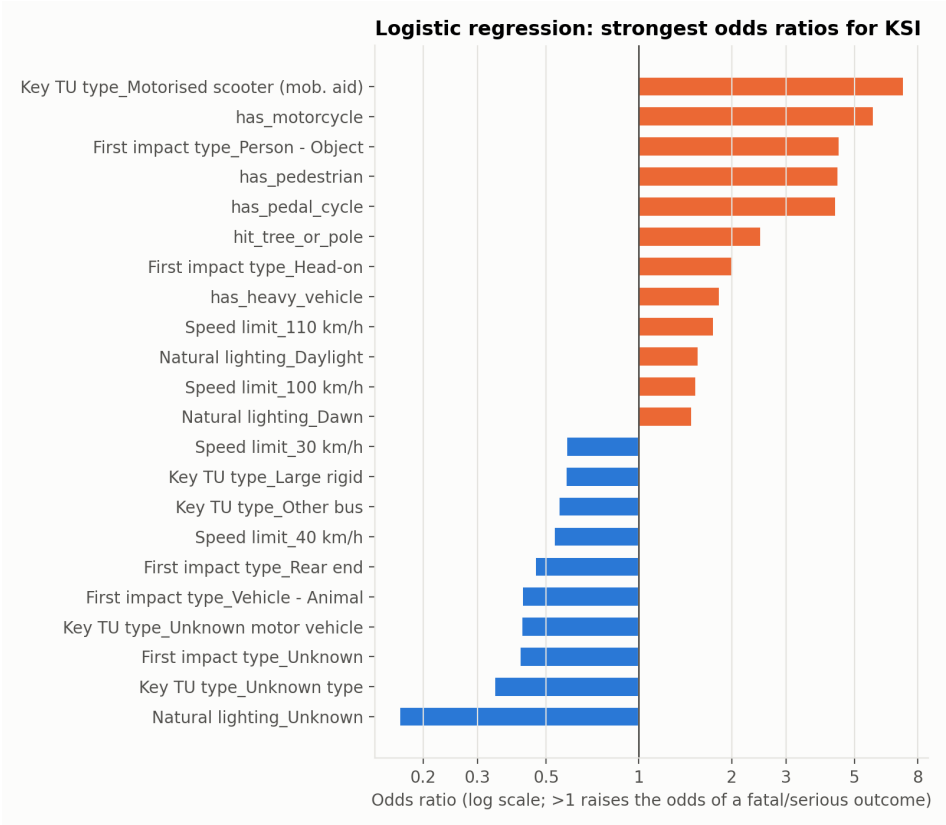


Figure 9: Strongest logistic-regression odds ratios for a KSI outcome (log scale).

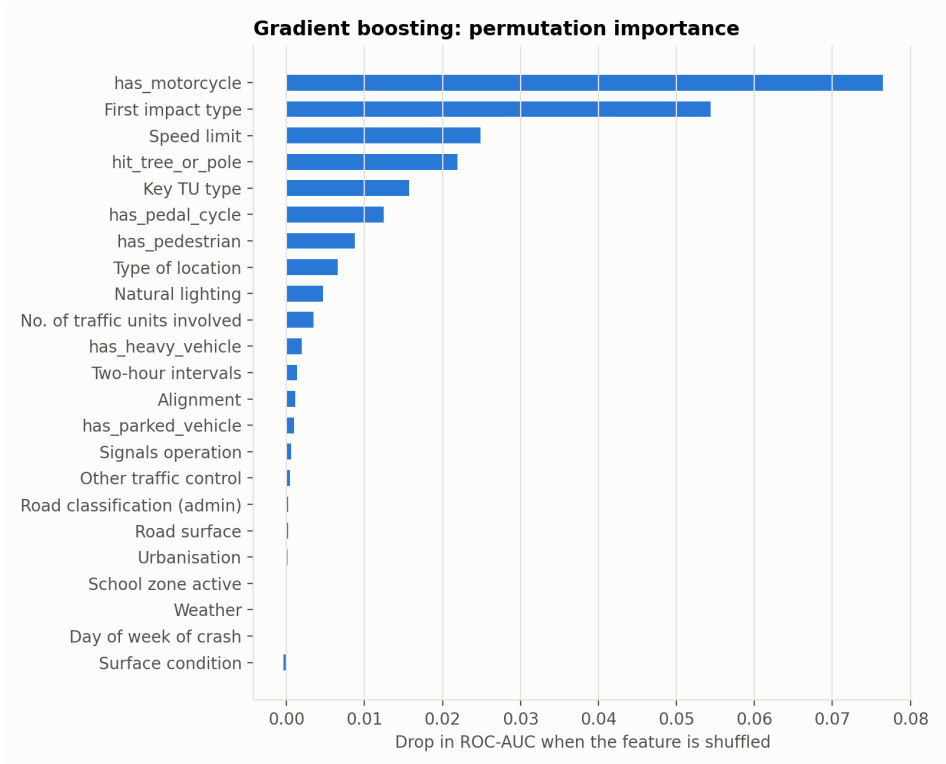


Figure 10: Gradient-boosting permutation importance (drop in test ROC-AUC when a feature is shuffled).

5.3 Interpretation and next steps

A ROC-AUC of about 0.75 is moderate, and the ceiling reflects the data rather than the model: the main determinants of injury severity, such as travel speed, restraint use, impairment, occupant age and vehicle safety rating, are not in the open release. Even so, the models separate low- and high-risk crash profiles clearly and give consistent, policy-relevant drivers. The next stage will:

1. recode structural blanks, treat the remaining "unknown" levels, and recalibrate probabilities;
2. add ABS population data to compare LGAs by per-capita KSI rate (Q3);
3. model the regional gap directly, controlling for speed, mechanism and vehicle mix (Q2);
4. apply density-based clustering to KSI crash locations to identify hotspots (Q3);
5. extend to the full ordered severity scale and a fatal-only model, and check stability with rolling-year validation.

References

Australian Bureau of Statistics. (2021). *Australian statistical geography standard (ASGS) Edition 3, July 2021–June 2026* [Licensed under CC BY 4.0]. ABS. https://www.abs.gov.au/statistics/standards/australian-statistical-geography-standard-asgs-edition-3/jul2021-jun2026

Breiman, L. (2001). Random forests. *Machine Learning*, *45*(1), 5–32. https://doi.org/10.1023/A:1010933404324

Carr, D. B., Littlefield, R. J., Nicholson, W. L., & Littlefield, J. S. (1987). Scatterplot matrix techniques for large N. *Journal of the American Statistical Association*, *82*(398), 424–436. https://doi.org/10.1080/01621459.1987.10478445

Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *The Annals of Statistics*, *29*(5), 1189–1232. https://doi.org/10.1214/aos/1013203451

He, H., & Garcia, E. A. (2009). Learning from imbalanced data. *IEEE Transactions on Knowledge and Data Engineering*, *21*(9), 1263–1284. https://doi.org/10.1109/TKDE.2008.239

Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, *9*(3), 90–95. https://doi.org/10.1109/MCSE.2007.55

Kaufman, S., Rosset, S., Perlich, C., & Stitelman, O. (2012). Leakage in data mining: Formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data*, *6*(4), 1–21. https://doi.org/10.1145/2382577.2382579

Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T.-Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems*, *30*, 3146–3154.

Mannering, F. L., & Bhat, C. R. (2014). Analytic methods in accident research: Methodological frontier and future directions. *Analytic Methods in Accident Research*, *1*, 1–22. https://doi.org/10.1016/j.amar.2013.09.001

McKinney, W. (2010). Data structures for statistical computing in Python. *Proceedings of the 9th Python in Science Conference*, 56–61. https://doi.org/10.25080/Majora-92bf1922-00a

Niculescu-Mizil, A., & Caruana, R. (2005). Predicting good probabilities with supervised learning. *Proceedings of the 22nd International Conference on Machine Learning*, 625–632. https://doi.org/10.1145/1102351.1102430

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, É. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, *12*, 2825–2830.

Saito, T., & Rehmsmeier, M. (2015). The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLOS ONE*, *10*(3), e0118432. https://doi.org/10.1371/journal.pone.0118432

Savolainen, P. T., Mannering, F. L., Lord, D., & Quddus, M. A. (2011). The statistical analysis of highway crash-injury severities: A review and assessment of methodological alternatives. *Accident Analysis & Prevention*, *43*(5), 1666–1676. https://doi.org/10.1016/j.aap.2011.03.025

Suster, C. (2025). *Nswgeo: Geospatial data and maps for New South Wales, Australia* (R package; Version 0.6.0.9000). https://github.com/cidm-ph/nswgeo

Transport for NSW. (2021, September). *Definitions and notes to support road crash data*. Centre for Road Safety, Transport for NSW. https://opendata.transport.nsw.gov.au/dataset/nsw-crash-data

Transport for NSW. (2022). *2026 road safety action plan: Toward zero trauma on NSW roads*. Transport for NSW. https://towardszero.nsw.gov.au/sites/default/files/2022-04/TNSW9659%20Road-Safety-Action-Plan-2026-ACC_1.pdf

Transport for NSW. (2025a, January 2). *Safer driving is the New Year's resolution NSW needs to lower road toll* [Media release]. Transport for NSW. https://www.transport.nsw.gov.au/system/files/media/documents/2025/250102%20MR%20Graham%20Aitchison%20annual%20road%20toll%202024.pdf

Transport for NSW. (2025b, October 9). *NSW crash data: NSW road crash data 2020–2024 (CRASH and TRAFFIC UNIT tables)* (Version 2020–2024 release) [Licensed under CC BY]. https://opendata.transport.nsw.gov.au/dataset/nsw-crash-data

World Health Organization. (2023). *Global status report on road safety 2023*. World Health Organization. https://www.who.int/publications/i/item/9789240086517